

## E.5 Fusion and stars

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: \_\_\_\_\_

Date: \_\_\_\_\_

Total: **26 marks**

1 pc = 3.26 ly. Show your working.

### Question 1 — The southern sky survey (7 marks)

An astronomy club measures the position of a nearby star in January and again in July, and finds an annual parallax angle of  $p = 0.25$  arcseconds.

- (a) Explain, with reference to the Earth's orbit, what stellar parallax is and why only nearby stars show a measurable one. [3]
- (b) Calculate the star's distance in parsecs, and convert it to light years. [3]
- (c) A member records the answer as "4 light years" by taking  $1/0.25$  and attaching the wrong unit. State the safeguard that prevents this slip. [1]

### Question 2 — The impostor twins (7 marks)

Stars X and Y look *equally bright* from Earth. Star X is three times as far away as star Y.

- (a) State the relationship between apparent brightness, luminosity and distance. [2]
- (b) Calculate the ratio  $L_X/L_Y$ . [3]
- (c) Explain why apparent brightness alone can never rank stars by power output. [2]

### Question 3 — The stellar seating chart (6 marks)

The Hertzsprung–Russell diagram seats every star according to two properties.

- (a) Name the two axes, and state the famously awkward convention on the temperature axis. [3]
- (b) State where on the diagram the following live: the main sequence; red giants; white dwarfs. [3]

### Question 4 — The reluctant protons (6 marks)

The Sun fuses hydrogen into helium in its core.

- (a) Explain why fusion requires enormous temperature and pressure, in terms of the forces between protons. [3]
- (b) The helium produced has slightly *less* mass than the hydrogen consumed. State what this mass difference becomes, and the equation governing it. [2]
- (c) State the property of the core that keeps a main-sequence star stable against its own gravity. [1]

## Answers

### Question 1 — The southern sky survey

- (a) As the Earth moves around its orbit, giving a baseline of 2 AU across six months, a nearby star appears to shift back and forth against the far more distant background stars. The parallax angle  $p$  is *half* the total annual shift. The angle shrinks as distance grows, so beyond roughly 100 pc it becomes smaller than the measurement uncertainty and the method fails.
- (b)  $d = \frac{1}{p} = \frac{1}{0.25} = 4.0$  pc, and  $4.0 \times 3.26 = 13$  ly.
- (c) The formula  $d = 1/p$  delivers *parsecs* by definition, provided  $p$  is in arcseconds. The member computed the right number, 4.0, and attached the wrong unit. The safeguard is to write the unit down with the formula before substituting, so that the conversion to light years is never skipped.

### Question 2 — The impostor twins

- (a)  $b = \frac{L}{4\pi d^2}$ : the apparent brightness is the luminosity spread out over a sphere of radius  $d$ .
- (b) Equal apparent brightness means  $L \propto d^2$ , so  $\frac{L_X}{L_Y} = 3^2 = 9$ . Star X is nine times as luminous.
- (c) Apparent brightness confounds a star's power output with its distance: a feeble nearby star and a brilliant remote one can look identical. Ranking by power therefore requires the distance to be measured independently first, for instance by parallax.

### Question 3 — The stellar seating chart

- (a) Vertical axis: *luminosity*, usually on a logarithmic scale and often in solar units. Horizontal axis: *surface temperature*. The awkward convention is that temperature *increases to the left*, so the axis runs backwards compared with every other graph students meet.
- (b) *Main sequence*: the broad diagonal band running from the top left, hot and luminous, down to the bottom right, cool and dim. *Red giants*: upper right, cool yet very luminous, which means they must be enormous. *White dwarfs*: lower left, hot yet very dim, which means they must be tiny.

### Question 4 — The reluctant protons

- (a) Protons carry like charges and repel electrostatically, the Coulomb barrier. To fuse they must approach within femtometres, where the strong nuclear force can take over and bind them. Only at enormous temperature, giving the protons sufficient speed, and enormous pressure, giving a high enough collision rate, do appreciable numbers get that close.
- (b) It becomes energy, carried away as radiation and as kinetic energy of the products, according to  $E = mc^2$ . For hydrogen fusing to helium the converted fraction is about 0.7% of the mass involved.
- (c) The outward radiation and thermal pressure generated by fusion in the core, which balances the inward pull of gravity. This balance is hydrostatic equilibrium, and it holds a main-sequence star stable for as long as its fuel lasts.